

ActInf GuestStream 066.1 ~ Adrian Bejan, "The Physics of Life: How to Predict Evolution"

Source: [YouTube](#) Playlist: GuestStreams Duration: 1:30:12 | Uploaded:
Unknown Transcript method: auto_caption

hello and welcome everyone this is active inference guest stream number 66.1 on December 6th 2023 today we are very excited and honored to have a presentation and discussion with Adrien Bon first in this session Susan Hasty is going to provide some slides and context Adrien will provide a talk and we'll have a discussion on through the hour and then Susan and I will stay on the live stream so if you have any comments or questions please feel free to write them so thank you both for joining and Susan to you thank you Daniel um and um and I'm so excited um and thank you so much uh for affording this opportunity Dr bang um um I thought that we would actually be again um uh I wanted to introduce you and as I was thinking about it I could spend probably the entire time just going over all the citations and awards that you've received and so I thought though that what I really wanted to share was um was something that is not necessarily in the public domain um I wanted to really dig into the significance of of the Constructor law um and the difference it's made um not only for I think the world which most people um don't appreciate I think and also the significance has made for me um it's definitely changing paradigms um across across the globe um this of course is um is Adrian's many uh many Publications and his many citations I think think there's 990,000 citations of his work alone um to date uh and um but what I'd like for the audience to really take away is um is really thinking about uh life in a new paradigm and how the Constructor law can inform us of um of how to think about a difference in not only just what is life uh but also what is power and what is evolution and change and and and and I'll give you a good example um of uh we we we will talk about a flow Paradigm a flow architecture and in thinking about architecture um one one idea that still permeates society and I think holds back Progressive how we think about um hierarchy and Adrian's books has definitely informed me um to think very differently you know we we still think about hierarchy um like we built them um in pyramids 4,500 years ago you know where we housed dead people and so this actually has a has the potential to change that and to really think

differently um one thing that a lot of people don't realize is is that um scientists from around the globe have been coming together for over 14 years annually um and presenting their findings and um and how they've taken this and spread it into every domain imaginable um this was actually a uh a doctor Dr Lucia who's I believe he's in Italy he shared with me just last night uh in talking about about it and how he's actually taken the constructal law and made Innovations in research in cancer and I thought his quote was just so powerful in terms of how we think differently about reality by really understanding and going very deep um and contemplating how the Constructor law um is is so significant um this actually what what I think is is um what is what I've discovered and uncovered um is is that it's actually spread far beyond the halls of Academia um and I'm going to make a bold prediction because we're going to be talking about predictions and I'm going to predict that uh Dr bang ban will be awarded the Nobel Prize because because it's not just the citations that of his work but when we look at and you go deep you're going to find that this the people are using the constructal law for example yeah this this one academic paper has already had um a many many citations in and of itself um you know it's being talked about in terms of religion and in terms of indigenous societies it's impacted architectures in how um airports and how structures are being made um and for me personally um I'm not coming out of Academia but but reading it and understanding The Constructor law really has given me an incredible all and appreciation for science it's actually given me a new appreciation for life um and and so here's just a few of the ways that um that chat GTP just put together here is how it's already spawned new ways of thinking about every area um of um of of application and Innovation uh and and so um what we're going to be talking about today is really focused on uh adrienne's books that are accessible to people like me I mean I would not been able to dig into the the many many Publications um and receive the the uh the depth of understanding of it without reading these books and I actually called them um um um the um the essence of strategic thinking because when we think about prediction we are thinking about how we're strategically going to move forward so today we're going to focus on the physics of life and Adrien has agreed to uh to give us another chance so we may not be able to cover all the many questions that have come through our Forum but we will be digging into them um after after he after we'll dig into some of them that come in on the chat today um and and then we will also be exploring them in a future in future session so so with that I'll turn it over to you Adrian and um and I look forward to this and I hope the audience really enjoys this and and as inspired as I've been see okay yeah thank you uh Susan and Daniel for the um invitation to to speak to your audience of

course I'm very grateful and humbled by the introduction um the um I thought that um in order to to get the discussion going it would help if I show you a few Concepts black on white and the slides I I happen to have at my disposal are from a very brief lecture I gave at the Third Constructor law conference in Ty in Italy just two months ago um it was a lecture about vascular flow structures but the the um the me of the um presentation is about how to predict the design Evolution uh vascular is in fact um a good title for the subject that interests your audience namely how the the brain works uh the brain is eminently a vascular architecture it is predictable from The Constructor law in in many ways um in many respects so um okay and so I um I start my presentation first of all with the definition of the meaning the meaning of the words I am I'm a professor meaning I teach students to uh um how to think and uh how to question themselves and how to to display humility humility when hearing ideas um and so the meaning of the word vascular from Latin is something that looks like um vessels conveying blood or lymph uh this is a a design an architecture that's uh that speaks of flow flow which means movement and hierarchy which as you'll see very soon is one kind of drawing and now now uh the word design should not scare anybody in who is in love with science design is poo pooed in physics um it shouldn't be because it really means the drawing drawing uh it came into English from French Desa uh the French got it from the Italians dieno it really means a sign an image that conveys a message and here I um have I just ja jaos the two versions of the same drawing a river Delta and the human lung to drive the point that the um the the drawing unites the animate with the inanimate and when the drawing does that it means that uh the phenomenon that the drawing represents Belongs To Nature not to a particular compartment in science or a particular department on the University campus uh the drawing unites uh and that is uh if there is one single message from my introduction here it is that um another thing about um um what I'm showing you is that natural things are physical physical they're observable visible palpable measurable and movable they are common sense they are not jargon I speak of drawings of architecture of images uh natural things have configuration shape structure Rhythm they have flow and and they their mother is the fact that they have freedom to change that means they have freedom to evolve and um um about the uh let's call it uh the freedom to evolve comes from the fact that these things are moving uh they're moving because they are set in motion by flow architectures that function as engines most most of these engines are natural only a few of them are human-made in any case um every single such engine is driven by one kind of energy source you can think about the sun driving the movement of the atmosphere the atmosphere is a Globe size engine the engine

produces power and the power drives movement and um while the movement uh is uh is the source of another phenomenon which is called dissipation or friction because the Mover has to get the environment out of the out of the way the environment the niche does not get out of the way unless it is pushed and in the rectangular box I show the engine driven by an energy source the work or power produced by the engine and then the fact that in the green brakes or interfaces between mover and environment are dissipating the power so if you sum up these white fat arrows uh the energy input from the energy source is dumped 100 100% into the environment but in the meantime Inside the Box because of flow architecture something was moved from one spot one place to another that phenomenon is called Locomotion it could be animal it could be vehicular it could be well current and all the other things we we know about so the bottom line is that nature is movement with low architecture that um unites really unites the geophysical with the biological with the social organization which is emotion uh these are images that change over time they are not the still art there are you think of them as movies many many movies running uh at the same time at the same time but every single one having the same plot the plot is that the movie has Direction meaning the changes have a directionality to them and I Rec recognize this uh published it first in 1996 several times then um it's called the construct the law of evolution in nature um or meaning the announcement that Hey listen it's the design in nature stupid uh has been overlooked in everywhere physics theonomics uh but not in engineering which is my uh my background um so I wrote that for a flow system to persist in time and that's the definition definition of the life state it must evolve with freedom such that it provides greater access I illustrate that with three drawings from the evolution of a river Bas on the floor of laboratory uh from time to time the um the architecture flows more and more easily because of erosion um and but at every uh point in time the the uh the tree sorry the the River Basin is a tree um The Constructor law announces probably screaming from the rooftops that there is a direction of uh of nature that points toward the future and that if you know this statement um if you run it through your through the wheels of your mind you have the ability to anticipate the so-called future uh to continue again because I teach the importance of knowing the meaning of the words and the importance of using Common Sense terminology here is what we mean by Evolution it is a sequence of changes in flow flow configuration in time freedom freedom is not some intangible it is actual physical features that allow a system to change and freedom can be measured uh in uh design um freedom is measured in terms of the number of degrees of freedom the number of ways in which you can change something uh Freedom can also be measured in terms of the effect of

increasing these this number of degrees of freedom has on the performance on the ability of the Mover to move more easily you can think of it as um uh okay straight jacketing the flow of a of a river well that River will flow uh um less fast meaning slower and um not as far as a river that's uh uh not straight jagged in fact the river that's straight jagged has a a uh a personality of its own to revolt and remove the obstacles out of the way the Fallen Tre log the levies at at New Orleans the history of the of of Earth is is is that it is about removing obstacles uh in order to secure greater freedom for what moves and this evolutionary design um as I just Illustrated facilitates access access to flow in areas volume volumes and that of course includes Society now access I checked the dictionary although I shouldn't have to because I come from a Latin based language it means the action of coming toward or near something it comes from aenus AED in in Latin it is a word available in English and this movement Naturally by itself untouched uh tends toward an architecture that's multiscale uh it's called hierarchy or arborescent if you will um it is a drawing um um that distinguishes itself through a few large channels uh cohabitating with many small channels in fact they're moving the same stuff and they're moving it in harmony the few are not exploiting the many uh the many uh are not revolting against the few they're all flowing together in order to facilitate flow over the entire available space and this configuration um Springs out of nowhere it is nature itself um my beginning in this um line of thinking was in actually it was before 1996 uh the problem that um U got got my publishing going is here on in the left left hand corner it was the question of how to shape a rectangular space that generates heat how to shape it so that the uh the generated Heat current um exits with greatest ease that means uh it is rejected by a rectangle that has the that is the the least hot meaning with the smallest temperature difference between itself and the ambient and in other words I gave myself one degree of Freedom which is the shape of the rectangle a shape that varied from Square to shallow rectangle the answer to my question was the U the the the rectangle in the in the middle which is not too shallow not too Square um but that was the click because from freedom to change something I discovered one drawing before this exercise when I posed the problem of how to shape it I not knowing the answer I did not have a drawing and you see in predicting Evolution the the thinking begins with a blank page it is a the most honest kind of beginning meaning the least biased way in which to start to start posing questions uh that uh that beginning is is a has a chapter a chapter with a blank page called the chapter is called I don't know you see I don't know but I discovered this and the rest is history and in um in the year 2000 I wrote the First full length book uh about the multitude of examples of this type ranging from geophysics river basins

lightning and everything snowflakes to u to physiology um breathing uh uh blood circulation ejaculation all these things are in in um in accord with with the Constructor law and um and then since then um I've shown a few other things that I will illustrate but the the the most important message from this beginning is that in science in science um quite often uh the first question is overlooked and the first question should be what is the phenomenon meaning what what what are we talking about and uh in the case of design in nature the phenomenon is meaning the elephant in the uh in the room is the evolution of flow configuration it could be the room it could be the field or or the ocean in other words ocean in other words nature itself so um that phenomenon is change with direction also known as uh uh the passing of time now uh getting closer to the subject of interest of these um um was these hierarchical architectures are can be simple or complex here I have uh two drawings of the same kind of flow it is a flow that connects one point with many many points on a on a circle um if I in the uh first uh design on the left the um the um the maker of the design had a very large number of degrees of freedom they are called L and and fi and Gamma and beta and X uh look and that took three years to do uh in the course of a doctoral thesis uh it's called The Perfect Design and then on the right one example that uh was done a lot faster by um by recognizing very few degrees of freedom called capital L and on the next Slide the performance of these two designs the perfect shown as black dots and the uh cheaper one uh with the white squares uh that performance is plotted with this lowercase f which is a number proportional to the flow resistance between the center and the perimeter and on the obsa this capital L is the number of points on the perimeter which means that the uh migrating to the right the design is more complex um you see here that um the two the two drawings are in fact uh they may look different even though they are both point to Circle trees uh but their performance is quite quite similar quite similar in other words a a simpler and cheaper design will do you see uh in other words the the maker of the design does not waste time the M of design is wise than the one that uh that polishes a um what was this sorry that gilds a the Lily you see um and um and that leads to uh to another uh way to organize the presentation which is that um if you if you polish something at infinum you're are not creating nothing so Perfection is the enemy of evolution this is a a dictum from volte originally it happens to be true from the point of view of the Constructor law um in fact uh I I discovered it but I also discovered the the first the first Discoverer in this Cas bter because I'm a fan of the history of good ideas and here's how the uh the eny the enemy Evolution if you know it uh benefits us uh the movement of people on the surface of the Earth is hierarchical we don't move around tracing the Grid on the surface of the

city we congregate We Gather uh we collect ourselves to one point of Attraction um use your imagination you know Public Square um and then at the end of the event we migrate outward to the area as a circular River delta and in so doing we organize our society hierarchically uh here is an even better example of how it is going because you see in the previous drawing which is the simplest possible this city block does not show the smallest alleys and the cucks and the uh The Lawns across which people walk but here in the uh drawing of the Atlanta airport uh we see the walking this is on left side vertically along the concourses um the uh the vehicle there's only one train shown as a black line um the by the way once again uh the the slow and the fast are carrying the same the same flow meaning the same crowd of people access is provided from any gate to any other gate and uh the figure of Merit where the performance of the hole I'm talking about the whole the whole flow system which is rectangular uh can be measured in several ways one of them is the average travel time uh between one gate and all the other Gates so keep that in mind travel time averaged over the area that's uh the measure of performance um the movement is with two speeds um fast along the train uh slower perpendicular to the train any one of us is is this capital letter q and um and the territory is has a is has a fixed size because area costs money but a variable shape a variable ratio of h ided l it is possible to show um almost on the back of an envelope uh how the um average travel time depends on the shape uh time is on the ordinate uh the shape is on the abissa and you see here a bucket shaped curve which has a minimum shown as a an empty circle uh the shape at that location is called $h / \text{by } L$ Optimum and the time is minimum at that in that particular design the um the discovery meaning to put physics uh at the disposal of vol is that a little bit of imperfection meaning uh travel time that's just slightly longer than the smallest imaginable uh give the gives the designer a significant bandwidth of um of freedom freedom to choose a shape that's okay comparable with the mathematically best shape but not exactly the same in fact the person who is empowered sorry endowed with this kind of secret uh will not spend time on finessing this particular SC sculpture to the point of identifying the the absolute best so imperfection imperfection or the the freedom the freedom of of choosing and then moving on is the mother of diversity in nature diversity happens from this freedom freedom not to be U uh not to fall in love with the absolute best and um there are people in several domains including biology who who talk about nature makes mistakes no nature doesn't make mistakes nature simply doesn't have time to to focus on on finessing itself uh what works is kept that is the slogan that I think uh uh comes closer to how nature Works uh one um aspect of um of movement of geophysical biological and also human is uh is the

so-called the logistics S curve um anything that spreads freely on an area or in a volume uh it spreads not only in space you can imagine the uh the milk spilled on the kitchen floor it spreads in time and the the invaded territory which in these Rising curves is indicated in the vertical Direction This invaded territory grows in S curve fashion it grows uh slow fast slow and uh the reason for that by the way here are a few more examples that are relevant to what I'm going to show you next uh these are ual measurements every single curve is an S curve on the left is the power production in the United States uh as an S curve um that is up to 10 years ago uh on the right are the total miles driven in the US uh per year and um at the very bottom are the the movers themselves meaning the world population that is also so an S curve which um um is already hitting the the ceiling the ceiling I remember when I was a an undergraduate in 1970 uh the rage was uh about the fact that there's a population explosion and uh the end of uh the end of oil and lots of other things of this kind um well um as I said at the beginning humility plays a role in science um it's much much more useful for The Thinker to start with I'm not so sure I'm not so sure let me look into it even if I believed these things uh it's very important that I qu I'm the I I I I am the first to question them and the reason why the S curve should be an S and it is predictable is this uh anything that spreads again the milk on the kitchen floor initially it invades the territory as something that moves fast and far and narrow like a finger and once the finger hits the the wall the wall uh the in this case the milk spreads laterally uh I call I call these two phases of of the occupation by using words from the military Invasion and then consolidation it's always this way and well but as you can see in the uh lower right hand corner um Invasion invades the area at an accelerate accelerating uh rate and consolidation continues at a decelerating rate and uh so these two arcs are spiced together obviously naturally and together they make up the S curve so this is the prediction the where does the um the hierarchy come in the hierarch hierarchy um is an addon meaning the next slide which I don't have uh which is that things that spread on a Surface spread in arborescent fashion not as a single needle as in this particular example and the the the the tree the tree spreads the stuff even faster than what I Illustrated here in other words something that spreads as a tree will have an S that that is taller during its uh um uh viral viral Intero you see and um so let's see that's basically the uh the um let's call it the the Preparatory uh material for for uh showing us who we are um the movement the movement of I this is an image of human movement today it is primarily primarily uh due to Aviation Aviation uh we all move on the globe from any point to any other point imaginable um Those Who U uh originate from let's say diametrically opposed points uh necessarily will have to flow through the heart of this

vascular architecture the heart is where this architecture was born which was uh Europe and North America uh and that is where the architecture has grown the most which is why it is in deep red right now and um so the movement is hierarchical the um the physics of movement is that to move more you must burn more fuel so the burning of fuel on the globe must be hierarchical meaning I'm looking at this image and I read the movement I'm another Adrian reads the burning of fuel and as you'll see next the burning of fuel is in fact proportional to the to the the to the wealth wealth available on the on the globe as in have money will travel and also on the next slide I'll show you that uh increasing hand inand with wealth and fuel use and movement is freedom and uh that's this is the final slide on the left hand side is the uh these are the facts these are data um people today like to uh to uh who pray to into in in the Church of data and data well here's data these are the facts uh those dots show that every group on the uh on the globe falls on the diagonal between energy use and an annual wealth which means that what is annual use of energy in physics en engineering is uh just another uh language uh in which to tell tell the story of of hierarchical distribution of wealth on the globe meaning that economics and and the physics are two sides of the same qu of the same coin meaning two ways in which to describe the same drawing and the drawing is hierarchical um all these dots are racing upward along the diagonal because as everybody knows every government reports annually uh its increase in GDP in in wealth well the government doesn't brag about the increase in the burning of fuel which of course means that no country is going to cut down uh its consumption of fuel um and on the right hand in other words the red arrow the direction of the bicycl and the pelaton in the two the France is the future on the right hand side the same data with the annual energy use on this time on the ordinate and on the obsa the the competitiveness with regard to economic freedom the the arrow that points toward the future is the same as on the left hand side the dots are the same as on the left hand side and they tell the story of a world uh that another p of bicyclists that are racing upward even though most most governments don't know it uh that the evolution of the Society of the globe is toward not only greater movement easier movement but also greater freedom and that takes me to the slide that Susan showed you um the um the the mother if I were to say the mother father of nature is uh in this direction from freedom of Change Is Possible from change evolution is possible and From Evolution the nature that we are and the which surrounds us um has become possible as well thank you thank you Susan would you like to provide a first reflection or Direction here um well before um it evaporates uh you were shaking your head so much Danel I want to see what's alive for you I wrote down many things I have many questions to

ask so but please feel free to ask one first and I'll prepar okay okay um what what really stood out for me um and I'm I'm slightly Prejudice because I've been going through the the physics of life so you that's uh the so many so many um ideas are captured in that but what really stands out for me Adrian is um is I'm seeing the system as object and structure as a verb no structure structure is a noun um structure is a um um it comes from the same word as well has the same origin as the word stun stun meaning uh Spread spread of in this case on a on a page or on a canvas uh it's a drawing it's a drawing with with a message streaming out of it so structure is um is a to structure is a verb clearly most most of uh this is now in the um uh in the languages of the indans tend to have both they could be both noun and verb that um is a is a feature of this group of languages but I don't think it's limited to the induran the point is that uh um yes I applaud everybody's interest in the meaning of the words because quite frankly um we are surrounded where we're actually drowning in jargon uh jargon is a good term it applies for uh it applies to uh locker room discussion I language that's understood only by 20 people all of them men you see um yes it's just like like any uh Department of physics oops sorry I didn't mean to put it that way but uh you get the idea uh the uh the object here which is why I continue to praise Galileo and D Vinci and these people the object is to to make these uh these ideas or AAS are accessible to uh to everybody to everybody it doesn't mean to dilute them or to dumb them not at all it means it means it means to display humility and to speak in a way not only to speak but to make drawings that uh that Open the Eyes of people and I'm fortunate because as a professor I have this freedom to in fact I just lectured today the freedom to make drawings the freedom to to we yes to tell a joke every once in a while um maybe a self-deprecating joke the point is that the uh uh the the flow of science the flow of science benefits from um from um first of all modesty but also the ability to uh ask questions and the to question the idea but to question the the speaker including yourself so so let me let me do that now um um and and help me understand where I'm confused because I guess I'm when I say the structure is verb I'm thinking about how we're talking about um configuration and reconfiguration and um and how the drawing actually turns into a movie oh that you know that I can get a you know my head around um and um and specifically you know what what I'm where I'm grappling with this and it's opening doors is how to think about you using the Constructor law the difference specifically between the tree structure which to me is shape and and the S curve and that is time so when I'm thinking about reconfiguration help me understand you where you when I'm thinking about movement and flow and the configuration the architecture um how you know how is that informing me well where am I well okay a um

a hierarchical drawing like the uh the River Basin uh has U many many shapes the biggest is the shape of the page inside of it there's the shape of the angle between two tributaries uh the drawing is full of such angles those angles are all different well but at the same time they're all similar because they're all kind of close to 75° or 50° um so you have shapes Galore if you if you if that is what interests you others others are interested in lengths well there are many channels and the the lengths are themselves hierarchical some are you know longer others are shorter um the um the more of these let's call them shapes and lengths uh the more complex the drawing is and complex you have to be very careful with this word is one that uh that um that places the discussion closer to the locker room meaning the the jargon because it hides hides these elements that I was uh enumerating uh complex the word itself means uh means Twisted together and um well that's if I use that word I I declare defeat not Victory you see if it's twisted together I don't even I cannot even wish myself good luck because there is no luck um in any case I know what it means and to me a word like that uh is an invitation to to question to question the word to question the speaker to ask myself what is the drawing that's come complex if you ask if you ask this question you discover that the drawing is not that complex because if if you can if you can make it it it is not not Twisted together you see it is Twisted together uh meaning um like one of these truly fractal drawings when you cannot draw it you see uh so on and on look we can we can uh let's say uh uh slice the orange here and small slices but we're wasting our time if we do that the point is that um to answer you the words that uh we use are very useful but uh they also tend to to u to to separate us to keep us apart if we do not agree on their meaning and that's why I interject Ed and I I added some imagery to the words that you speaking okay and by the way structure same origin as stun well that's where the scurve phenomenon is you know the the spreading is as curve uh that's growth also known as growth um the collecting is also as curved uh which is also growth you know during the the the the reaping of harvest the the flow of the grain from the area to The Silo is a stream that grows slow fast slow um the direction of the flow is not the phenomenon the history of the flow is the phenomenon and it is called the S curve um also known as the logistics uh S curve or the logistics curve and and um I can tell you the history of that if you're interested the point is that Logistics come from a military military um what's the word um yeah they call Logistics the the flow of material uh food not just ammunition but uh mobile hospitals and the infirmaries and all these things that are flowing and moving uh behind behind the troops um and um there's a great quote from Napoleon Napoleon bonapart U he said uh um amateur he was talking about generals amateurs talk talk strategy professionals talk

Logistics Logistics and uh that was Napoleon uh the quote is even better known because of the English language as belonging to General Omar Bradley uh who used it without crediting Napoleon so so but in any case these are two very important figures who knew uh which end is up think about it highly successful uh because you see it today in in the conflicts that are raging it's not about what goes on not meaning not only what goes on on the on the front it is about the um Logistics where the the the the the the power of the society behind behind the um the uh the military activity this is extremely important to uh to to to learn from every uh news item today well thank you I'll give a reflection and connect just a few areas of flow that that I think really speak to some of the points so the key reflection is nature is not broken by Department it's not broken or mistaken in any way and that is radically different than many organizational structures including in learning and education and so that's a big question mark in a big open space and then there was some pictures and some words and some symbols but that's a starting place that's perennial um I thought of two kind of flows evoked in your description the first was drawing with a ballpoint pen on the page and the ink as it diffuses through the paper and about how there's a trace and then there's you could imagine filling the channels of the paper and different properties of the paper and just about how the act of drawing is and and why the line fixes at a certain point that's the S curve has flowed to its flow it dri solvent is gone so on so you can go into more detail but just in that act and then another one perhaps more in a informational space is the jargon and the jargon can be piercing but it can be a grappling hook to go 100 miles out and get really specific I'm talking about Theta sub Delta of figure four of your 2012 paper and that can go all the way down and then there's that second phase of expansion it's not just about throwing needles it's about throwing the needle to the wall that was your engineered setting and then if it makes contact then there's this secondary phase that's the actual Overture and the initial precision technical detail is a first Contact if even that and again that gives Science Education and practice I think a real Challenge and opportunity uh I think you hit the the nail on the head three three times real hard and it's the same uh is the same nail uh even though every time you it made the issued a different screen um the um but that is in fact very accurate very accurate and it happens just a second and and it does happen the way that you uh you professed uh it does happen even after it hits the wall in all your examples because because happening is nature itself um the U the the let's say the campus divided into departments that don't talk to each other is unnatural it's unnatural uh these barriers have been uh uh disregarded by by clear by clear thinkers uh all the time the ones who have made progress or contributions are the professors or researchers that

had the innate ability to to uh to to scale fences and jump over them or to get drunk and step off the curb and fall on the street uh these are the people who made the history of science uh they the name they are called polymaths okay and U we have them I know even here at Duke I know the the few who are who are this way they are my friends and um and uh we're we're too busy to start a revolution but the the future is uh the way the way you you sketched it it will be it will continue to be one of filling in the spaces or rediscovering or repeating what others have done in the sketchy fashion or maybe throwing a jargon word out there or perhaps as a joke as a joke or or even uh even more effective as a non PC joke and then people say ah Aha and then and um in the beginning the this act this act the this spark you you described um takes roots and fills the space because because a good idea is like food for the hungry it attracts it attracts participants it attracts students it attracts disciples followers second generation teachers or artists and I'll come back to your first example when when you started speaking I saw immediately the history of painting the first thing that DaVinci and these people Michelangelo did was to make a a very fast pencil sketch something that only they recognized because it was already in their minds and today with all this instrumentation people are discovering as traces of graffiti on the on the canvas that had been covered much later with with various colors over and over again yes that's the way that's the way nature really works and um and nature is also the brain itself let's face it there's no other there's no other architecture that works except the one that I've have sketched crudely in my uh in my uh Exposition here Susan do you want to give a kind of closing question or comment while we have Adrien I'm I'm gonna begin with a blank slate I I'm I'm I'm I'm available a little longer if you if you have a food for the hungry here oh I'll ask a question from the live chat if if all right yeah upcycle Club writes a question to Professor Bon what are the biggest unanswered questions or areas of exploration within constructal theory that you hope future researchers will investigate good question well uh the I've been kicking this ball um for um what almost 40 years um not uh not in the direction of discovering uh the Biggers that need to be discovered or to be explored but out of curiosity today as youve saw in Susan's presentation there's a world movement it's populated primarily by by younger people who are in fact competing with me uh clearly I'm I'm not at the start of my career but they are beginning and in fact they're quite uh creative and uh productive um the um my own uh new work is about uh uh asking questions that about this field that have not been asked or answered uh the constructed law has not been refuted by anyone in its uh you know already four decades nobody so I decided to question it myself to say listen let me uh what was that word let me um uh uh let me prove it wrong but there's this word from Carl

popper about falsification yeah yeah let me let me falsify okay and so I I'm looking for phenomena that at first uh viewing seemed to contradict the Constructor law yes and uh because I have the talent of looking around and uh and um yeah I keep my eyes open yes uh when there's a heavy downpour and the the rain falls on um on dirt the um the the obviously the river River Basin is born but the river channels uh flow um with cataract they develop a periodic um sequence of little pools in inter space with the manyi waterfalls cataract yes the river is uh putting the brakes on itself yes it's a contradiction of the constructed law right so I questioned it and I published it and I discovered in the course of U uh proving The Constructor law wrong The Constructor law is right the water that flows in a periodic fashion flows faster than uh in the textbook of water flowing uniformly and steadily down an incline of a along a inclined plane uh and the reason the reason why the interrupted flow is faster is there's a there's nothing nothing more um perfect for the flow of water than to fall without friction to fall over a dam so the water the water well the water and the erodable soil have in them in themselves this tendency called freedom freedom to facilitate access and I uh I uh I was yes I was excited to not that not to prove myself right I was actually trying to find the um limited applicability of this of this concept uh and in this case I did not find it and I'm continuing this sort of I'm um I'm the okay in the same rum I'm both you know prophet and G fly I'm trying to prove prove Adrian wrong okay and uh it's exciting but uh I'm a I'm an audience of one uh I talk to my wife and she knows where I'm coming from and uh all of it is worth is worth laughing in fact but if we don't laugh uh we uh we have no future frankly so just to kind of wrap up the um I think I read the paper you're talking about but I remember uh something about the edes um and cor and yeah and that kind of brought me back to what you know what Daniel's question was about is far as you know how we break free these silos um and you how does how does that how does that form um and also I think that also um speaks to maybe the second session that we can hold which is about the freedom because you we really that is of itself that's a topic that is just so rich I'm happy you mention the edies because that was an earlier um um example of uh U my questioning this this principle people all over people all over view turbulence especially in in the applied World turbulence is something bad something that increases friction um yes yes something that's um not as good as the smooth flow called laminar flow uh the one with the smooth streamlines and this is what you see in the text yeah turbulence this complicated thing you know edies rubbing and well you know what turbulent flow is constructal as well because because compared with a lamina turbulent flow uh uh answers the question of what really flows What flows is not the fluid What flows is

the movement across the fluid meaning the movement from the fast to the slow in other words the mixing the mixing is the mixing is known in physics as the transfer of momentum uh that is what's facilitated by this St change in the drawing from smooth uh layers to to things that appear to be complicated but by the way they're not complicated when I did predict the sizes of these edes and their speeds and their hierarchy they are hierarchical the edies the edies are basically these trees the trees that I drew uh like the Trees of a River Basin uh but the the the River Basin that becomes this turbulent pool is a is a is a is a tree the channels of which are embedded in in the in the same fluid which which is the the the pool and uh every CH every channel uh is unstable and becomes curved becomes an Edie and it rolls over and um the the uh the the tree that I just sketched uh becomes a bouquet a bouquet of broken branches that continue to roll they continue to roll filling the space but you can see their birth in in the little speech that I gave um and in any case there's a hierarchy of edes in every so-called turbulent flow field now the doctrine of turbulence research is not like what I just spoke it is about a a uniform distribution over the field over the pool of EDS of various sizes as if there are uh uh you know fruits of various sizes all thrown into a pot and mixed around with with the chocolate sauce you know uh that's not what turbulence is turbulence is a is a broken tree structure the branches of which keep rolling uh embedded in each other and uh and that that so-called mixture is the one that Vehicles the movement of movement across the the volume without it without it um without it there would be no uh frankly without turbulence there would be no life on Earth uh everything would be stagnant pool of I don't know what a layer a layer of no life uh and then another layer of no life and a you know in other words no mixing the by the way so mixing we know this from other stories of Science and History mixing is in fact um the mother of not only life that's you get species from uh from from um events of of encounter you know encounter or meeting um and um new species from for example from inter breathing and so on and um so without mixing there's nothing on Earth nothing so one final question um that speaks back to your slide on nothing moves unless it's pushed yeah that's right that's is if one us to explain explain that uh yeah a peasant knows that the horse is needed don't need only a card um and I come back to uh through uh the news of the day the um um the the logistics the the ability to move things the ability to power things this is uh this is Extreme I would say is the most important because Society itself is alive because of the of the power that it the society produces and that the power by the way something else that people don't realize is spent as soon as it is produced it is spent and and destroyed dissipated as soon as it is made and that is called movement or life life means to destroy this the power

that is available to you you don't have to look at today's Advanced Society look at any animal any animal um the movement of the animal is the physical movie of how the U the um useful energy embodied in the in the food is being destroyed uh no food no animal movement simple as that no food and no um engine which is the animal itself that means death not life death and uh these are basic facts of physics you don't need an education and whatever anthropology or uh yeah you could use an an education in a veterinarian medicine that's where you get this much broader picture of Who We Are um thank you Adrien we really appreciated it hope to talk to you another time all right thank you thank you for the opportunity to uh get to know each other thank you yes thank you so much very insightful thank you well Susan I have many random remarks to share would you like to share anything you go ahead go ahead and of course what's going on in the in the chat yeah all right I'll I'll read a few um traveling fast and slow that reminded me of thinking fast and slow and then in the airport there was the fast backbone and the slow walking on the edges but in terms of thinking fast and slow actually the fast part is on the edges of the biological organ ISM with the reflexes and the reactivity and the slower part is the backbone so it's like kind of the same but different um another piece was he said that the first question is often overlooked like what phenomena are we looking at what's the system of interest and then that made me think about like biomechanics like an elbow moving and so in the physics textbook or in a example it's like well there's a weight on the end of a stiff rod and it's set up but that's kind of The Last Mile question the phenomena of the arm is a broader phenomena and so you still may want to have that very reduced disciplinary model it's small it's useful it can give insight and so on it's just part of a bigger tapestry about that phenomena and that's a different starting point for the conversation and collaboration on that topic that's not disciplinary so the entry point of the real system and the entry and returning point is by definition the broader conversation than the kind of terminal scientific artifact which as we discussed that's kind of the narrow but far as opposed to the broader and and kind of softer but more space enclosing yes and actually that brings up um you when you think about phenomenon um yeah I almost want to go to you know what is the context but phenomenon is actually you know what is Flowing what you know what so when we kind of pull that in as far as the Fast and the slow how much and you know a lot of times by injecting that you're actually putting turbulence into a conversation which you know which if somebody is um associating turbulence with conflict then they might be actually distorting it extorting the freedom of what can emerge um and so you know this comes back to the you know Invasion acceleration he spoke about and then the consolidation is the deceleration yeah

a few few points on that first I think a question I'd love to learn more about is like Trends are not monotonic so just because things have been a certain way for the last 50 or 100 years doesn't even mean the sign is going to be the same so I'd be interested to see how that kind of change comes into play but I'm thinking about a cars engine there's Parts where the laminar flow is the desirable trait for transfer and then there's Parts where mixing like liquid and gas and thermal mixing is more important and they simply are not the functional they play complimentary functional roles it's literally like fast and slow and there's never a reason why especially considered apart from a specific system it basically can't be said which of those Tendencies are you know even you know how to compare them which is why even though we were looking at a two-dimensional shape with the rectangle there was only one degree of Freedom which was the ratio so that reflects I think having a simple we need to capture that yeah yeah capture that yeah this this actually so if we were to translate this and you bring in some active inference you what's kind of coming alive for me in this is the difference between exploration and exploitation yeah like he showed a a parabola with a ball y rolling to the bottom of the hill and then there was kind of a a zone of tolerance like within 10% of the Y AIS and so it pretty shallow bowl wide Bowl so there was a lot of play that would keep it within a 10% Tolerance on the white now to persistently optimize that all the ball needs to know is the slope of which direction it should head it given that it's already knowing that it's within this zone of control ability which is how all bodies are everyone has you know been part of an unbroken chain of blood sugar regulation so the evolutionary and the developmental prior is that the physiological state is already in the zone of control and then it's just a directional issue to head towards the bottom of the bowl but you can't stop X Change You Can't Stop climate from changing you can't stop physiology from changing that's not to say you can't drisk and think about real consequences of action but just taking it at the most narrow technical sense of change through time and evolution all things are changing there's no point in trying to Halt change and it's a simpler actionable heris for the ball on the ball uh bottom of the bowl which way is the free energy flowing yep have the least resistance um and and you this kind of comes back to the phenomenon of how do we predict um and um and in fact I would suggest let me share my screen because for those people listening um yes I just uh go for it see yeah do you see it so yeah I mean just putting that you being able I I const I constantly go back to and reread this law in fact the reason I'd said earlier that I'm going back to a clean slate because it was like you know whenever I whenever I have the opportunity to speak with Adrian he puts something out there and I'm like so what I thought you I was

reading into it it was kind of yeah that's kind of going to that the phenomenon of exploration um and you know seeing what's out there you know how are we perceiving the world right um and um and then how do we switch to exploitation and when I'm saying exploitation I'm talking about how do we move from being in the world to how do we move to remaking the world or or or changing you doing what he talked about originally is is how do we um how do we construct um the the um the world around us for you know for our Niche construction actually he does reference Niche Construction in the book um and uh and so how do we you how do we do that so that we're using the least amount of necessary energy because I think that one question that I had which I think would be a a um topic for a future conversation is is how do we know when we are um violating the Constructor law yeah and this ties back to the belief updating if I could I'd like to connect a few of the words you have on this slide to the ACM ontology and that's kind of like as we explore all the time that's how we kind of begin that bidirectional conversation so like for a finite size system that is a thing system of interest it's a thing it's the blanket and the internal states of the particular partition the particle the thing itself to persist in time non equilibrium steady state or a similar phrasing talking about the path of least action of the stationary State or the attractor or the trajectory or the path of that thing MH so we're talking about the same kind of system of interest and then evolve with freedom as we heard these terms are plain and so evolve being changed Through Time notably not taking on the kind of intergenerational connotation of evolution some people like in biology would say something like well Evolution happens between gener generations and development and ecology happens within a generation but here time evolution is being used more generally Freedom was described in terms of degrees of freedom statistics degrees of freedom and we could also think about rule-based degrees of freedom as affordances as the knobs that can be changed those are the degrees of freedom on that interface and then to the last piece the providing easier and greater access to What flows here I um think to the recent guest stream 63.1 that discussed the philosophical status of material and information flows and moral Logistics yes and then of course the the the joke I guess was like the logistic curve but it's like logistic and Logistics and then what status to Accord to material flows and informational Flows In integrated autopoietic mortal computational systems where there's a weaving of like the blood flow of the brain and the ion flow and the nutrients and there's an informational flow semantic flow that is not strictly described by the material architecture now now to pull back to to okay so put that in the frame of reference in terms of one thing he said that was like blew my mind was um it's not the it's not the what is flowing when he was referencing what is

flowing and I'm you know and so what you just said um where I'm going with this and you if this if tell me if if this is where you're thinking is epistemic and pragmatic value so what is Flowing I mean once we we can calculate that and he also gave a a calculation for degrees of freedom nice these are great directions I'll I'll just go to a different angle I thought it was I loved his recurrent connection with drawing and art and about the sketches that are being discovered behind paintings that's like deeper blankets of the painting deeper levels of historical meaning and cross cross referencing like oh this one is signed at an earlier date and we have this letter so we know that they were speaking with this person so it's like deeper in space and time and causality and then even our artist like Kandinsky would go through many iterations and many sketches on even nonrepresentational or post-representational however you want to think about the kind of art that's not like a landscape but the painting was then synesthetic with the auditory and so that kind of brings together not to resolve but for the beauty of the snapshot the noun structure and the Dynamics to structure as a verb and as a process and like the fact that you can't disentangle or drop one or the other prioritize one or the other they're literally snapshots of the movie Yes yes and and to kind of tie that back to um you the path of least action um and but it reminds me of a saying and correct me correct if correct me if I'm not getting this correct but yeah the difference between um the saying goes is the difference between an average artist and a and a expert or a master artist is knowing when to paint the last paint stroke that's cool it reminds me of I would have written a shorter letter if I had more time like there's these kind of are they intuitive or are they counterintuitive because it's knowing what not to do and when to do it yeah yes yes and this yeah this yes yeah that's that's a that's a great uh way to uh to wrap that up yeah because it comes back to you know the the the path of least resistance and you know and the and the freedom and so often times the affordances if people don't understand that you they're actually pushing too hard or overdesigning um and you how do we how do we use the Constructor law and active inference to to to his point I mean it's like power is spent as soon as it's produced I'll make a few more contact points with active inference one question I have is like where's the Observer we've explored a lot this year with Chris fields's course in the quantum information sciences and so on about the role of the Observer so should we think of these flow processes as only or actually occurring on the material substrate if so where is the space for the Observer or the poly Computing of Mike so that's one open question and then I think another important touch point is like Notions of hierarchy which he returned to and there was like a very unconventional hierarchy represented with people's diffusion from city blocks like smaller and

bigger parties getting going to and leaving at different times and then the diffusion Maps across the city and so it's like that's one representation he kind of contrasted that flow hierarchy with like a pyramid um type representation maybe you mentioned this and then that made me think about two ways that we see the hierarchical models in active inference one of them that are different than both the pyramid and the um city blocks one of them is the spatial enclosure nested blankets nested models that are enclosing each other not just diffusing into the same city and then another is metacognitive with the word of saned Smith at all in live stream 28 with the hierarchical model representing metacognitive layers so between the pyramid the city streets the enclosure and the mind that one symbol Bridges us already such that it puts the Bruns of the question on what to do and how to do it yeah yeah lot so much to expl makes that's exciting uh so to wrap up and to capture is there any other questions not that we have to answer them now but if you can you just kind of read off any other questions that might have shown up in the live chat I'll capture those and you know and we can start kind of a repository of what you know what showed up today um and uh and that way we can use it to explore explore further yeah we will continue the kind of series and the curating of questions I'll just read one Mills 8102 wrote what are the factors or conditions that afford the restructuring of a flow system does it always require the imposition of sufficient Demand on that system to change does it take a crisis great question and one one question I have is where are perception action do you have any last comments or thoughts just my head spinning wow well wow great thank you Susan and thank you farewell yes till next time absolutely bye